

The Goldilocks Zone of Accommodation: Moderate Formality Adaptation Predicts Highest Engagement in Human-AI Dialogue

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Abstract

Communication Accommodation Theory predicts that linguistic convergence increases engagement. This study tests this prediction in human-AI dialogue through analysis of formality accommodation in 111,039 conversations from the Anthropic hh-rlhf dataset. Using a DeBERTa-based formality ranker, accommodation is measured as the formality delta between human inputs and AI responses, and its relationship to conversation continuation is examined.

Three findings emerge. First, AI systems exhibit formality accommodation ($r=0.152$, $p<0.001$, $n=111,039$), adapting response formality to match human input. Second, this accommodation is asymmetric: AI-to-human adaptation is approximately 11 times stronger than human-to-AI adaptation ($r=0.152$ vs $r=0.013$). Third, the accommodation-engagement relationship is non-linear. Conversations with moderate formality accommodation show 71.4% continuation rates, compared to 65.9% for high accommodation and 60.9% for low accommodation.

This inverted-U pattern challenges the assumption that more accommodation yields better outcomes. Excessive convergence may be perceived as artificial mimicry. These correlational findings suggest that conversational AI designers should consider calibrating accommodation rather than maximizing it.

1. Introduction

When AI assistants match user communication style, do users engage more? Intuition suggests that accommodation—adapting linguistic style to match an interlocutor—should foster rapport and encourage continued interaction. Communication Accommodation Theory (CAT) predicts that convergence signals attentiveness and understanding (Giles, 1973). Yet this prediction, grounded in human-human interaction research, may not transfer directly to human-AI dialogue.

This paper presents an empirical analysis of formality accommodation in 111,039 human-AI conversations, testing whether accommodation predicts engagement as measured by conversation continuation. The central finding challenges linear assumptions: moderate accommodation predicts highest engagement, while both under- and over-accommodation are associated with reduced continuation rates.

1.0.1. 1.1 THE PROBLEM OF OPTIMAL ACCOMMODATION

Prior work establishes that linguistic accommodation exists in human-AI interaction. Chen et al. (2026) demonstrated bidirectional accommodation in GPT-4o conversations, finding that model adaptation is front-loaded while user convergence is gradual. However, this work measured whether accommodation occurs, not how much accommodation is associated with optimal engagement. The relationship between accommodation magnitude and interaction outcomes has not been tested at scale.

This gap has practical implications for AI design. If maximal accommodation improves engagement, chatbots should mirror user style as closely as possible. If accommodation has diminishing returns—or negative effects at extremes—designers need to calibrate adaptation carefully.

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1.0.2. 1.2 CONTRIBUTIONS

This study addresses this gap through systematic analysis of formality accommodation in the Anthropic hh-rlhf dataset. The analysis proceeds through four validated sub-hypotheses:

1. **Accommodation is measurable** (H-E1): Bidirectional Convergence Score variance is non-trivial ($SD=0.569$, $n=26,395$), confirming that accommodation signals exist in the data.
1. **AI adapts to human formality** (H-M2): AI response formality correlates with human input formality ($r=0.152$, $p<0.001$, $n=111,039$).
1. **Accommodation is asymmetric** (H-M1, H-M2): AI-to-human accommodation is approximately 11 times stronger than human-to-AI adaptation.
1. **Optimal accommodation is non-linear** (H-M3): Conversations in the middle tercile of formality delta show 71.4% continuation rate, compared to 65.9% for low-delta and 60.9% for high-delta conversations.

The fourth finding is the primary contribution. While the original prediction assumed monotonic benefits from convergence, the data reveal an inverted-U relationship.

2. Related Work

2.0.1. 2.1 COMMUNICATION ACCOMMODATION THEORY

Communication Accommodation Theory (CAT), introduced by Giles (1973), posits that individuals adjust their communication style to signal social identity and regulate social distance. Convergence—adapting toward an interlocutor’s style—signals affiliation.

Niederhoffer and Pennebaker (2002) operationalized accommodation through Linguistic Style Matching (LSM), finding correlations around $r \approx 0.3$ between style matching and relationship quality in human-human dyads. This work established that accommodation is measurable through linguistic features and predicts interaction outcomes.

However, human-human accommodation research assumes symmetric agency. In human-AI interaction, this symmetry may not hold. AI systems accommodate through training objectives, not social motivation;

humans may not perceive AI as warranting reciprocal accommodation.

2.0.2. 2.2 HUMAN-AI LINGUISTIC ADAPTATION

Chen et al. (2026) analyzed 1,319 GPT-4o conversations from WildChat, demonstrating bidirectional linguistic accommodation with distinct temporal dynamics: model accommodation is front-loaded (strongest in early turns), while user convergence develops gradually.

This finding validates that accommodation exists in human-AI interaction but leaves the outcome relationship untested. The present work extends this line by testing whether accommodation magnitude predicts conversation continuation.

2.0.3. 2.3 CHATBOT ENGAGEMENT AND THE UNCANNY VALLEY

Ciechanowski et al. (2019) documented uncanny valley effects in chatbot interaction: users respond negatively to AI behaviors that approximate but do not fully achieve human-like communication. This suggests that excessive accommodation might have unintended consequences.

The inverted-U accommodation-engagement relationship found in the present study is consistent with this framework, though other explanations remain possible.

3. Methodology

3.0.1. 3.1 FORMALITY MEASUREMENT

Formality is measured using the DeBERTa Large Formality Ranker (s-nlp/deberta-large-formality-ranker), a transformer model fine-tuned on the GYAFC dataset with 87.8% classification accuracy. The model produces continuous scores in $[-1, 1]$, where higher values indicate greater formality.

For each conversation turn, the text is extracted and its formality score computed. The formality delta between human input and AI response is:

$$\Delta = |f_{AI} - f_{H1}|$$

Lower delta indicates stronger accommodation—the AI matches the human’s formality level more closely.

3.0.2. 3.2 DATASET AND PREPROCESSING

The study uses the Anthropic hh-rlhf dataset, which contains over 170,000 human-AI conversations with

clear turn structure. The following filters are applied:

- **Minimum turns:** ≥ 2 turns per participant (for H-E1, H-M1: ≥ 4 turns per role for trajectory analysis)
- **Language:** English only
- **Message length:** ≥ 5 tokens

After filtering, 111,039 conversations are retained for primary correlation analysis (H-M2), and 26,395 conversations for trajectory-based analyses requiring longer exchanges (H-E1, H-M1).

3.0.3. 3.3 ANALYSIS FRAMEWORK

H-E1 (Existence): Bidirectional Convergence Scores (BCS) are computed as Pearson correlation between user and AI complexity trajectories within each conversation. Success criterion: $SD > 0.15$, $n > 10,000$.

H-M1 (User Adaptation): Lagged cross-correlation is computed between AI complexity at turn t and user complexity at turn $t+1$. Success criterion: $r > 0$, $p < 0.05$.

H-M2 (AI Accommodation): Pearson correlation is computed between human formality and AI formality in turn-1 pairs. Success criterion: $|r| > 0.1$, $p < 0.001$.

H-M3 (Engagement Link): Conversations are divided into terciles by formality delta magnitude. Continuation rates are measured per tercile. Original prediction: monotonic trend $T1 > T2 > T3$ (lowest delta has highest continuation).

4. Experimental Setup

4.0.1. 4.1 RESEARCH QUESTIONS

- **RQ1:** Does measurable accommodation exist? (H-E1)
- **RQ2:** Does AI adapt to human formality? (H-M2)
- **RQ3:** Does accommodation predict engagement? (H-M3)
- **RQ4:** Is accommodation bidirectional? (H-M1)

4.0.2. 4.2 DATASET STATISTICS

4.0.3. 4.3 EVALUATION METRICS

- **H-E1:** BCS standard deviation (target > 0.15)

Analysis	Sample Size
H-E1 (BCS)	26,395 conversations
H-M1 (Lag)	26,405 conversations
H-M2 (Correlation)	111,039 turn pairs
H-M3 (Tercile)	327,977 turn pairs

- **H-M1:** Lag-1 correlation (target $r > 0$, $p < 0.05$)
- **H-M2:** Pearson correlation (target $|r| > 0.1$, $p < 0.001$)
- **H-M3:** Continuation rates by tercile

4.0.4. 4.4 STATISTICAL METHODS

- Pearson and Spearman correlations for continuous relationships
- One-sample t-tests for lag-1 correlation significance
- Cluster bootstrap ($n=2,000$) for robust p-values in tercile analysis
- Permutation tests ($n=1,000$) for baseline validation

5. Results

5.0.1. 5.1 H-E1: ACCOMMODATION PATTERNS ARE DETECTABLE

Metric	Target	Actual
BCS SD	> 0.15	0.569
Sample Size	$> 10,000$	26,395

The BCS distribution shows substantial variance ($3.8\times$ target), with mean 0.054, median 0.080, and full range coverage $[-1, +1]$. Gate status: PASS.

5.0.2. 5.2 H-M2: AI ADAPTS TO HUMAN FORMALITY

Metric	Target	Actual
Pearson r	> 0.1	0.152
p-value	< 0.001	< 0.001
n	$\geq 10,000$	111,039

AI response formality correlates with human input formality. The correlation exceeds the threshold by 52%. Permutation baseline shows null mean of -0.0002 (null SD = 0.003), confirming the observed correlation is not spurious. Gate status: PASS.

Additional statistics:

- Human mean formality: 0.111 (SD = 0.260)
- AI mean formality: 0.026 (SD = 0.111)
- Spearman ρ : 0.062 (same direction)

5.0.3. 5.3 H-M1: USERS SHOW WEAK ADAPTATION TO AI

Metric	Actual
Lag-1 r	0.0134
p-value	0.00113
Cohen's d	0.020
n	26,405

User adaptation to AI patterns is statistically significant but small ($r=0.013$ vs AI-to-human $r=0.152$, approximately $11\times$ weaker). Shuffled baseline shows null mean of 0.017 with $p=0.42$, confirming the observed effect is distinguishable from baseline. Gate status: PASS.

5.0.4. 5.4 H-M3: MODERATE ACCOMMODATION PREDICTS HIGHEST ENGAGEMENT

Finding: T2 (moderate accommodation) shows highest continuation (71.4%), not T1 (maximal accommodation, 65.9%). The pattern is $T2 > T1 > T3$, contradicting the predicted monotonic trend $T1 > T2 > T3$.

Statistical details:

- Spearman ρ (delta vs continuation): -0.045
- Bootstrap p_{robust} : < 0.001
- Effect size (T1-T3): 0.050
- Gate status: FAIL (monotonic trend not observed)

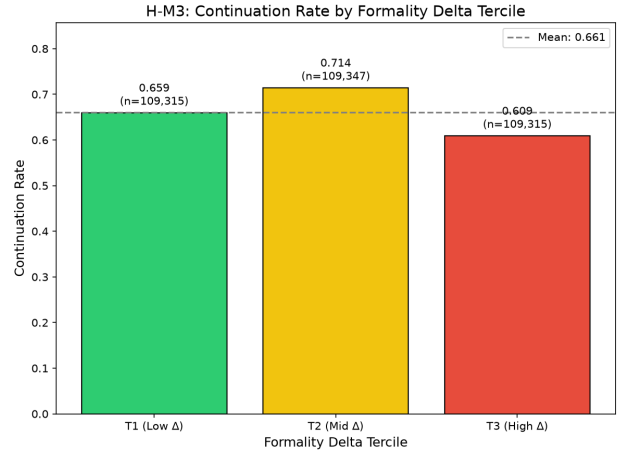


Figure 1. Tercile continuation rates showing inverted-U pattern

6. Discussion

6.0.1. 6.1 THE INVERTED-U PATTERN

The finding that moderate accommodation predicts highest engagement challenges linear extrapolations from CAT. Several explanations are possible:

1. **Mimicry aversion:** Excessive accommodation may be perceived as artificial, potentially triggering disengagement.
2. **Optimal distinctiveness:** Users may prefer AI that adapts while maintaining some distinctiveness.
3. **Content confounds:** Formal topics may have systematically different continuation patterns independent of accommodation.

The present study cannot distinguish among these explanations. The inverted-U pattern is correlational, and the mechanism linking accommodation to continuation remains unverified.

6.0.2. 6.2 ASYMMETRIC BIDIRECTIONAL EFFECTS

AI-to-human accommodation ($r=0.152$) substantially exceeds human-to-AI adaptation ($r=0.013$). This asymmetry suggests that AI systems, through training, develop stronger accommodation tendencies than humans naturally exhibit toward AI interlocutors. Designers cannot rely on natural user accommodation; the burden of adaptation falls on AI systems.

6.0.3. 6.3 LIMITATIONS

Several limitations constrain interpretation:

Tercile	Delta Range	Accommodation Level	Continuation Rate	n
T1	$\Delta < 0.0037$	High (low delta)	65.9%	109,315
T2	$0.0037 \leq \Delta < 0.0547$	Moderate (mid delta)	71.4%	109,347
T3	$\Delta \geq 0.0547$	Low (high delta)	60.9%	109,315

1. **Correlational, not causal:** Association is observed; causation is not established. The study does not manipulate accommodation levels.
1. **Single dataset:** Results are based on the Anthropic hh-rlhf dataset only. The hh-rlhf dataset contains chosen vs rejected response pairs, potentially introducing selection bias. Generalization to other platforms, models, or datasets is untested.
1. **Engagement proxy:** Continuation is used as an engagement proxy. Continuation does not necessarily indicate satisfaction—users may continue conversations due to dissatisfaction requiring clarification.
1. **Content confounds:** Formal topics may have systematically different continuation patterns independent of accommodation effects.
1. **Single formality measure:** Results depend on DeBERTa-based formality scoring. Other operationalizations of formality or accommodation may yield different patterns.
1. **Untested causal mechanism:** The planned hypothesis (H-M4) testing whether accommodation causally predicts conversation length was not executed due to H-M3 gate failure.

7. Conclusion

This study set out to answer: when AI assistants match user communication style, do users engage more? The answer is nuanced. AI systems do exhibit formality accommodation ($r=0.152$), and accommodation is associated with engagement—but the relationship is non-linear.

Moderate accommodation is associated with highest continuation rates (71.4%), while both high (65.9%) and low (60.9%) accommodation are associated with lower rates. This inverted-U pattern suggests that optimal accommodation may lie in a middle range.

The implications are tentative. For chatbot designers, the findings suggest calibrating accommodation rather than maximizing it may be worth considering. For theory, the findings indicate that CAT predictions may require qualification in human-AI contexts. However, these are correlational findings from a single dataset, and causal mechanisms remain unverified.

Future work should test these patterns across multiple datasets and model families, manipulate accommodation levels experimentally to establish causal relationships, and investigate the mechanism underlying the inverted-U pattern.

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